

Michael Levin & Franz Kuchling ~ Active Inference Insights 015 ~ Morphogenesis, Selfhood, Teleonomy

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hello everyone and welcome back to active inference insights I'm your host darus P way and today I feel extremely lucky to be able to interview two exceptional researchers Dr Michael Levin is a distinguished professor of biology at T University serving as director of the Allen Discovery Center at tus and the tus Center for regenerative and developmental biology his lab studies anatomical and behavioral decision- making at multiple scales of biological artificial and hybrid systems fr cooking is Mike's postto at toughs with a special interest in the physics of life and how cells adapt to changes in their physical environment guys welcome to the show thank you so much for joining me it's an absolute pleasure and an honor um you're kind of you know everyone in active inference has a foray into theoretical biology but I would say you guys are the first out andout biologist that we've had on the show so I'm super excited to learn um but you may have to excuse some of my more Layman ideas that come from the British schooling system where I wanted to start is probably just putting in some groundwork for the audience uh who might be unaware of certain terms that get used in in the papers that you guys write I wanted to start with morphogenesis what is morphogenesis and in what kind of you talk about moros space what is moros space and how do those two things relate yeah I don't know who want to take it I guess I I'll start and and then France can give his view um so so thanks for having us um it's a pleasure to to be here and to talk to you um uh there there's a there's a fundamental fact of biology which is that uh complex shape and and functional structure are self-assembled so uh let's just think about embryonic development we all Start Life as a single cell and eventually that single cell has to give rise to a a a complex organism of different types and so this not only during development but also during regeneration so for example a salamander loses a limb those cells have to not only uh restore the entire structure but then also know to stop when it's done okay and uh there are other examples of this of course metamorphosis when caterpillar gets to a butterfly or or something like that so there there are numerous examples where the biological Hardware is uh is is basically

reconstructing itself and so this is broadly conceived this is the notion of morphogenesis now one way to look at this is as a sort of open loop process meaning that there are a set of rules that are executed and parallel so um lots of molecule cells and so on are following these rules and then uh there's a process of emergence and something complex comes out the other end you know this is the standard story of cellular automata or other ways to think about emergent complexity and that's true I mean that does happen and that's certainly part of the story but the more interesting part of the story is that morphogenesis whether developmental regenerative or in the context of cancer suppression or remodeling any of those kinds of um examples one of the amazing things that it can do is get to its Target morphology despite various perturbations so if you want to dig in I'll give you I can give you all kinds of interesting examples where not only the environment changes not only injury but actually the components themselves the the amount of DNA the type of DNA the number of cells the size of cells all of these things can change and certain morphogenetic systems are amazing at getting where they're going despite all of these things in fact they will often take U novel paths to do it which is uh William James' definition of intelligence right same goal by different means so um one way to so so so then so then the question is what what conceptual tools can we bring to bear to understand this process uh and what we have settled on or or at least what what we've been we've been uh pursuing for the last some some years is this idea of navigation that what's actually happening here is that you have a collective intelligence it's basically a set of processes um hierarchically organized from molecules to cells to tissues and so on and what they're all doing is navigating the space of all possible outcomes so there's a there's a huge um you know space of of diverse morphologies that they can achieve and what they're doing is like any agent navigating its environment they are they are trying to get to where they're going and sometimes they have interesting competencies to do it even when things go wrong so now we can bring to bear all of the tools of behavior science of of control theory all this you know cybernetics all the stuff that's used to design um autonomous vehicles you know all of these kinds of things to ask how does the system know where it is how does it remember where it's going and what competencies does it have to get there so that's that's that's why we use the notion of morphospace because it allows us to make testable specific hypotheses about the policies that that are involved in this process wonderful that's a that's a really nice background from which we can kind of explore these ideas yeah it seems when you put it that way very consonant with the the ideas of Act of inference which is obviously something that you guys have have written about before so friend maybe I'll come to you now you there's this idea in

these so so people think of priors in sort of statistical space or more broadly about the organism as a whole you know what are the priors of the organism to be welfare to eat and so on but we could also talk about a cell's prior or an a cell's posterior and basian unfolding basian processes within the cell what in this idea of navigation and self organization what would you say the cells prior is is it purely to reach a kind of Target morphology or is there anything more than that I would say there's a lot more than that but that's the most um the simplest to kind of formulate because morphospace like Mike was um explaining is something that we we have much more access to as a development of biologist right so the the challenge is you can formulate all kinds of Prior and belief expectations for your but if you have no way to to estimating how that's even being achieved and that's hard to really um put in any theory that that's you know experimentally accessible but for one thing kind to point out that the idea of a morphospace or cells navigating uh spaces is is only challenging for for people that not in involved in this field because we so used in spatial Dimension as that being the space right but you can assign all kinds of Dimensions to to a cell we do this for humans all the time you know the idea of of characters in in a plot being multidimensional is something that is very much applicable to celles as well so to ask to answer your question more recently so what what happens in terms of expectations of a cell it might have you have to really go into what is it constantly achieving has any relevance for future States so that's La is can be transcriptional States so what kind of genes are being expressed at any point in time and that is being informed not just by the by the structure of the DNA so by millions of years of evolution it's also been formed by past um events not just in in that individual cell but in it parent cells and and and going on so that's what we call the epigenetic landscape so which genes are actually even accessible for the transcription machinery and which genes are being activated to really transcribe and make certain proteins down the line and so that really is something where where a lot of the pr expectations are going into um that we at least now in the lab are really investigating of what kind of transcription profiles do we see but then also in terms of this it's membrane potential you know and lab studies bi potentials a lot that's something that of course for Neuroscience much more familiar we talk a lot about Action potentials we don't have those in in non- neuronal cells but every cell has a has a m potential electric potential and those are informative and they are um causing lot of Downstream events so we can get more downt online but that's part of what I think is um part the PRI belief and expectations that sell is constantly having excellent yeah um I think there's a really important point there which is kind of something that I noticed reading your papers which is that

morphogenesis the the the very idea of physical space seems to be prioritized um it's self-organization in space is that the only way just in terms of conceptualization is that the only way that we could think of a cell you know reaching a posterior that aligns of its prior does it have to be in accordance with some kind of self-organization space or is there another way that a cell could kind of reach a homeostatic set point I'm absolutely it doesn't have to be just space right so um even if you have a cell just somewhere in the body already differentiated already kind of has its morphology achieved it has to maintain that morphology first of all but then also it has to perform a certain function right your kidney cells your liver cells they have very different functions even though they're all fixed in space and physical space so PRI beliefs are you know in this case for example for kidney cell is what kind of um what kind of um changes in the blood are being seen I being exposed to Parts in the liver same thing um if you have cells in in the brain why it's different they have expectations about Action potentials so um it that space right and again morphospace isn't just um spatial Dimension right it's also it's also traversing because every cell body this an important point every cell in your body has exact same DNA more or less right but what's actually being expressed that's the the difference between all of them so they are vastly different in the actual proteins but also in the genes are being expressed constantly so when we say traversing and navigating the morphospace we really talk a lot not just about that physical space but how are they changing which genes are being expressed between being turn on and off which really makes the cell identity in the end excellent yeah yeah that that's that's absolutely really really useful to integrate function and space there and I guess those two they feel to me inextricable but actually perhaps so I'm wrong there Mike is it how so so reading kind of the active influence account of morphogenesis we talk about the external states of the mark of blanket for a cell being other cells and in the case of Developmental processes you it turns they end up being other specific type of cells that they then join together with in morphospace is that how necessary is the presence of specific other cells in the basin unfold for a particular cell or could you imagine that a cell's prior might be confirmed you know absent a specific collection of external cells yeah I mean this is this is a great question and it's it's part of a very emerging research program because it really isn't super clear uh yet uh what precisely cells are paying attention to and and what their internal models of the outside world are what we do know is that in for example let's just take let's just take early development so you have a blastoderm and there are I don't know 50,000 cells there and at some point we as external observers will look at that and we'll say well there's an embryo there's one embryo well what is there one of right and and if you if

you want to think about an embryo the thing you're really counting when you say one is you're counting the fact that all of the cells have bought into the same picture of the outside world and what their goal is so so they are all going to cooperate to build a particular thing what binds them together into being one larger scale emergent individual is commitment to a particular model of of what where where they um where they belong where the boundary is between them and and some other embryo opar because you can you can make by putting in little little scratches into that blastoderm you can make twins triplets you know there are multiple individuals that can come from that Blaster so every cell um has to be part of some in in that scenario has to be part of some Network and that network has to make decisions about where it ends and where the outside world begins because that has to do with size control and and you know Left Right asymmetry and all kinds of things so uh it's really critical to understand what exactly um the the cells are paying attention to what exactly the uh tissue is paying attention to so the cell network because I actually think that there are uh there are agents here at every level so even actually even to take a step back you know um even molecular networks within cells I don't even think you have to get to the to the level of a cell before you can start talking about um beliefs and and expectations and so on because as we see even Gene regulatory networks can learn from experience they they can do something like five or six different kinds of learning including pavlovian conditioning and so you can you can ask that question about what does the world look like from the perspective of a gene regulatory Network before it's been trained or after it's been trained and so on right so this is all of course very very active uh areas of Investigation still so we don't know all the answers but but I think I think it's very important to start taking the perspective of the things that we are modeling and asking what do they see and what what are they measure ing what are they paying attention to how do they remap information into Salient um compressions for them you know and and and how that affects decision- making and so on how does that then um integrate with a kind of idea of Telos or tonomy um because you know the way you if we take the perspective of a cell or a collection of cells that are parametrizing beliefs over external States it sounds purposive it sounds like as you say there's agency there um and even the word beliefs people associate with propositional philosophical beliefs how far should we take that metaphor literally well okay so so the first thing I would say about that is just as kind of a ground ground rules I I don't believe in the dichotomy of literally versus metaphorically okay I think all we have in science is metaphor and and and what what you know when people talk about molecular Pathways I mean talk about a metaphor right the pathway relative to what's

actually going I mean this is completely metaphorical so all of all of these things are metaphors the real the real question of course is what set of tools does that metaphor give you to make progress what does it help you discover what does it facilitate so uh to me all of this is gets settled empirically to the to the uh to the point that uh if if you have a particular metaphor so maybe you talk about cells as having beliefs or maybe you don't and the real question is what does that worldview allow you to do and I think that's the final judge of it I don't I don't really put a lot of weight into sort of philosophical pre-commitments about what what goals and beliefs have to be it's a question of to Port portability of tools so um I now now having said that so so I think that uh these kind of um uh these kind of tools are extremely useful and they're again not binary because even even as early as the 40s already uh uh weer Rosen Bluth and Bigalow gave us a um kind of a scale of different types of different goals of different complexity right so it's so very simple kinds of passive matter and then active matter and then and homeostat various kinds of homeostats and second order metacognition and so you can you can sort of go up and up and and it's an empirical question which of these are the most useful for any of the contexts that you study but I think it is uh it is absolutely reasonable to see all of these things as having some position on that scale and you know it's important to say that back back before I don't know let's let's say before the the the the 30s and 40s there was a the the problem the problem with theology was that uh we didn't have a way of thinking about machines with goals we sort of Mo most people would assume that that humans have goals but other than that we really had no idea how to cash that out but but it's it's really critical to to say that we're past that now we we have a good rigorous science of uh what it means for for mechanisms to have goals it's fine it's not magical anymore we don't have to be be be afraid of it and um and so uh I guess guess the last thing I want to say is the difference between thology and tonomy so um the def one definition of tonomy is apparent gold directedness okay a parent thology and a lot of times people use that as a kind of um as a hedge as a way of softening the concept they say well look it's not real thology it just looks like Tey you know and and that's a way so so we don't have to really um uh commit to be to being teist we're just going to say just looks that way so so I like the word tonomy but for the opposite reason I am not trying to soften the concept I'm I I I'm you know not in in in no way uh um kind of hesitant about using tileology I think it absolutely Works what I think is uh useful about the concept of tonomy is that by putting the word apparent in there what it does is it focuses on an observer's point of view apparent to whom and I think that is a really critical question to ask in almost all of the stuff that that we talk about here which is not some some sort of objective uh uh

Universal uh truth about these things but from the perspective of some particular Observer now that might be a human scientist that might be a parasite that might be a subcomponent of the system that might be some sort of superet of the system that might be the system itself right we all you know systems observe themselves too and so so so what that says is that there is there is some perspective from which the system acts as a goal directed agent and thus that could and then can be can be used with tools that belong to that so um that's that's what you know that's that's what I think is the the most useful term of uh the sense of tonomy yeah France I was wondering what you thought about this because um I think it's a natural inclination maybe it's a romantic fiction that we want to ascribe some teos to smaller let's say or or less complex entities like cells or processes within cells there's an interesting thing that Mike said that I wanted to pick up on which is that basically everything in science is a metaphor I guess where I might slightly push P back on that is the science of lived experience um and phenomenology which we can't help we have to take in some ways literally because it's the one thing that we're intimately aware of and there's a lovely paper by Anthony Jack and Andreas rof ages ago 2002 I think where they argue that basically you can take sort of every psychological construct and say well every the way that we you know coales on what attention means or memory means really is because of what we how we experience it so I was wondering whether there's a kind of uh tie there I mean or association between when we talk about a cell's beliefs a cell's attention a cell's memory is there any way we can think of that without bringing in our own phenomenal experience it's tough I mean um like especially initially when I earlier in my my research when I you know wasn't as deep into a lot of the the metaphysics behind it and all that I try to really avoid it and I try to really focus the my audience on on just the mechanical aspects mechanistic aspects you know the cell aspects but I found that it was almost impossible to do that because all the words we use are so loaded with everything we have from psychology um but that being said nowadays um I I think really the important part is to do to to do that transversion transversing those two um fields and both ways because the the the claim we're making I think is not that like a seller is doing the same thing on the same level on the same scale like you know Mike really mentioned right there's there's a certain gradient there as a as a human being is but we're trying to understand where the origin of all these phenomena are coming from and are there and and more importantly are the fundamental principles shared between itself and uh and into more classical active INF agent like a human being so things like beliefs right so when you say that S as a belief people like oh well you talking about you know um having religious beliefs and then you get really

fastly trapped into into these loopholes but I nowadays try to counter this argument by saying well what is a belief and like where does it come from it can't just it has to a human being has to come from certain expectations right so you really have to then go down to how how do you once he has to Define you have to then go to some mathematical Frameworks eventually otherwise it's tough to do without it and so active inference is nice because there you have very precise definitions of what a belief actually is yeah and so once the kind of trick you can do is keep a w okay let's talk about Act of inference human being need you make a certain process you know you have a general model General process you define the mark of blanket so separation between what's outside that you can't see and then yourself so an internal State and then you have a separation of those two by the active and sensory inputs you can get right you have sensory inputs and then you can act upon the environment to change what those sensory inputs your eyes your ears are are perceiving over time that's actually what agency really is about it's about controlling you can't really control the environment if you can't perceive it what you're actually controlling is how you perceive the environment but of course you're doing that by acting on the environment that's what agency is sure now once you put it in such a fastic framework then the I think transferring that to sales becomes much more much more easier and much less um controversial I think yeah I I I sense that that sort of um less philosophically Laden more deflationary account of agency is is definitely possible it's just control in line with prior expectations or or you know self organiz principles of self-organization I think one thing that perhaps Act of inference also hasn't got very clear on but certainly isn't clear in Psychology is selfhood and in some ways I feel like it's very fundamental to this question of morphogenesis and self-organization in biological space because it's morphogenesis for who right so likee you sort of noted that in many way well we're talking about that we have it's all Observer relative but also in some ways it's also internal relative because they are man you know they are self organizing for the sake of their own Integrity so well again that might be ascribing a Telos that's unwarranted but it it could be described like that um what is biological selfhood and I guess for people who think of the self as one monolithic thing how does it how does it be how can it be distinguished from more complex forms of selfhood like metacognitive selfhood or epistemic selfhood and so on um I think that uh you know the issue of morphogenesis for whom is actually really important uh because you know a a really defective frog embryo might be a very nice zebra fish embryo and so this whole right and so and so this whole notion of You Know M mutation and uh and just just in general evolutionary change and what what exactly is a birth defect is really

important and interesting chemistry doesn't make mistakes so there's no such thing as a mistake in chemistry but there is potentially a mistake in carrying out developmental biology depending on what perspective you're looking at so it's something very interesting is you know the the ability to make mistakes is a Hallmark of a cognitive system right because because that suggests that you had certain expectations that were not met and so on so just the very nature of a of a birth defect is is interesting so now now the notion of a self uh I of course there are many ways to think about it I'll give you what how how I've been thinking about it um what one of the things that that I really wanted to do uh early on is to come up with a framework uh to directly think and compare think about and compare agents of very diverse structure and Providence so we're talking about you know conventional beings like like us and animals I mean also those are also of course Collective intelligences right we're all made of cells and so on but also you know new synthetic things that are going to be created um robots AI hybs h ibds cyborgs you know all kinds of mixed kinds of things potential exobiological life and so on what what do they all have in common you know what what do all agents have in common no matter how they're made and no matter how they got here you know what their origin story is so what what I settled on and and so now you see you know I'm super into the whole thology thing so so what I settled on is this notion of um the size of their goals in other words uh I I Define this notion of a cognitive likee which is um it's not the dis that you can sense or act it's the size of the goals that you're able to maintain the biggest goal that you could possibly maintain is the size of your cognitive likeon so you know so so every every agent has some degree of memory going backwards maybe large maybe small some predictive power forward and some spatial area of concern over which its goals play out and um and so so now now the thing about that is that that is the the determination of a cognitive lyone by an observer for example here you as a scientists are given a new system you don't really know how it works and you want to do some behavioral experiments and you find out that here are the things that it can possibly care about that that is a a third-person investigation of its agency and we say this is an agent to what what I think of as a useful definition of the self is exactly that process in first person so a self is what how that process plays out for the system itself so the system itself also has to be able to corar grain the outside world and not only tell agential stories about um things in its environment that do things right because no no system can afford to to track micro States you know lassus demon isn't going to you you'll be eaten and and dead long before you calculate anything so so every every agent in the real world has to has to course grain and they they tell these stories about you know agents other agents doing things

but you also have to tell that story about yourself to some extent you can't avoid it because if you don't have a compact uh self model you are not going to be very good at controlling your own Parts in making things happen it's just right that's just part of being effective in the real world and so I think what we say when we um what we mean when we say self is the internal first-person view of a system about its own right so so so so you so so there's kind of a bundle of things you need to have you need to have a sense of boundary between you and the outside world so which are the things that I think are actually me versus versus not me you need to have some representation of some space that you think you're operating in you need to have some sense of your own sensors and effectors what what can I actually do what am I actually sensing and so on we as third person Observer scientists uh try to tell those stories about other things that we work with but but agents tell them about themselves and so do par you know parasites and comp specifics and various you know potential mates and prey and cheaters and and all these things everybody's hacking everybody else and in order to hack other other agents to hack yourself and your own parts to have control over your own Parts requires that sense of selfhood and it might be you know as as France said the point isn't to say that every Rock has human level hopes and dreams that isn't the point the point is that uh there there's a huge range in how much processing is how much sort of meta processing is done about all of this you know you can you can roll all this forward in a in a very kind of minimal way or you can have all kinds of Loops where you're actually monitoring your own progress on certain goals and you have the ability to change your goals and to pick new goals I mean there's a you know you can layer stuff on and on but that's yeah that's really great that's a really lovely explanation yeah it really deeply aligns with the way that I think about selford which is fundamentally from a sort of mearian um phenomenal self- modeling aspect and that's exactly what Thomas would say which is that self-modeling is the capacity to recapitulate or resimulate the actual data structures of your own body in a kind of Co grain space so it's you know the the one thing that came to mind is talking about the uh cognitive light cone is in terms of the the extent of your goal yes it's in space but it's also in time and that feels like a very relevant thing here um when I've spoken to Carl and and when I've seen the direction of active inference it's very much going in time uh in in you know the direction of temporal planning deep temporal planning which arguably relates to the depth of your generative hierarchy because the deeper your generative hierarchy you know the the highest levels are going to be tracking the most invariant and long-term fluctuations in the external environment so friend I'm just curious if you're I mean this is just a thought experiment is there a way

where perhaps if you have a cyborg or a tadpole or you know some entity that you're examining and you're trying to get to its cognitive infrastructure is looking I don't know whether this would be behaviorally or anatomically is looking at its capacity for planning which might well just be its the depth of its generative hierarchy does that give us an insight into its actual you know existential or or experiential life I think it does and it relates back to the your early question about what are its actual beliefs you know what on what level are its beliefs being expressed and so I just gave you a couple examples what I left out is you know that of course like you just said what's important is the hierarchy of that how are these different different beliefs organized in a certain hierarchy of of spatial and temporal scale and also to to I really like the point Mike earlier by chemistry not making mistakes right so what that means is the conclusion of that idea is that any purely chemical system is not going to change much it's become entirely predictable it's not going to push back on anything in the environment it's going to have a certain it's going to come to an equilibrium pretty pretty fast usually and then that's that's it and so another way to form this question of how does a cell how can we um test agency to answer your question how how we test um having a cell certain certain ability to form hierarchy and and form beliefs we have to understand of how much push back can it can it give you can it give to the environment over over long times and so the way I would probe this hypothetical cyborg or or tadpole in terms of like what kind of um expectations what kind of planning is it is it doing is you have to really see how these different parts of that generative hierarchical model are interacting so you will have a you have a quick first response which could be you know purely physical but then very often on turns very active in most system that's the the active inference part is really what makes life life take you know it's it is entirely active and it's active very quickly but also over long time scales and so um what we're trying to show in this one paper by recognition with Chris fields and Mike was that that's a drive that got selected for very early on in in evolution and so the way to Pro the system is really looking having different reporters um generically but also just having certain dies in A system that will give you readouts of its its active States like what is it making what kind of Poli is making is it changing it's I of potential and then you try to relate those things are they completely independent are they just you know a non non living system you expect those to be pretty pretty static with certain very predictable Dynamic um reoccurrences of patterns but in the living systems they're going to change and they're going to change dependently on the scale so one scale is going to affect the other one and it's going to continue to doing that and that's I think how the planning is achieved by that temporal in uh independent inter

interdependence between those different aspects of a cell if they were independent they would be no planning but if they're dependent of on each other then the cell is basic going to be using its long-term processes to redirect and inform it its active points at any moment in time nice yeah yeah yeah and speaks to a kind of almost predictive coding hierarchy um where you have these kind of trickle down predictions Mike did you want to add yeah just I just want to go back back for a second to something you said before which I think is quite interesting um about the uh the you know the the the non-metaphorical nature of our phenomenal experience and I I I I take your point and I think phenomenal experience is really important but but there's there's the issue that we change right and so and so how you inter so so I mean for us so let's say puberty is a is a is an example right where where a lot of veence changes and things that you thought were really important suddenly not important at all but all this other stuff is now important and the the example that I really find uh kind of in meaningful for all this is the caterpillar to butterfly transition so what happens there is you've got this you've got this softbed creature that has a that lives in a two-dimensional world and it crawls around and it has a particular controller for that kind of soft body you know actuation so it's got a certain kind of brain and then it has to be transformed into a completely different creature that's a hard-bodied kind of vehicle it flies it lives in a three-dimensional world and so on and so so it has a completely different brain and during that that process the brain is basically mostly broken down most of the cells are killed off um and then reassembled you know it grows it grows a new brain so so there's there's sort of three levels of interesting things here the the first level is that that that aspect of phenomenal experience like what's it never mind what's it like to be a butterfly but what's it like to be a caterpillar becoming a butterfly and how much of right and how much of the um the metaphors that how much of the things that the caterpillar took very seriously the butterfly thinks were total metaphors at this point right if it could I mean I'm not saying that's how it thinks about these things but but right things that were quite um quite concrete for the for the for the caterpillar are no longer so for the butterfly that's the first thing the second thing is that um we know there's continuity you know if you if you like um the stability of memories as a Criterion for selfhood which a lot of philosophers like uh we know that the butterfly remembers things that the caterpillar learned so there is some continuity here even though the the brain cognitive system in the body of all of that stuff is radically changed but there's some continuity and so the next kind of question you think about is okay where the heck is this memory where when the that survives total brain re you know refactoring but the third the third thing that I find even even more important about all of this is

is comes from the simple fact that um the way the way the way they test these memories is well one one way to do it is you um you you feed the caterpillar leaves on a particular color disc and then what you find is that the so it forms the Association and then you find that the butterfly goes back to that disc to feed right but the but the key the key thing here is that butterflies and caterpillars don't eat the same stuff right the caterpillar eats leaves the butterfly doesn't care about leaves the butterfly wants nectar so um so now something very important you cannot just you know Persistence of of of the self in terms of memories what you can't do is just keep the memories constant because you'll lose the salience they make no sense to those memories will make no sense to the butterfly and in fact what you have to do something very interesting you can't learn leaves on this color you have to generalize to the to the to the higher category of food and so right and and and then you have to remap so this is the part that I'm super interested about in terms of um moving memories around and and and transplants of memories and you know personal identity and all that you have to take the the Deep lessons of your past life and sort of re uh reinterpret them for your new life which is by the way completely different anatomy and all of that but there are some things you learn that are still pretty useful just not exactly in the way right it's not the it's not the details that that matter now it's the kind of the generic concept that matters so so re remapping right remapping that information onto a new exper onto a a a new um embodiment for for for that for that being that's that's why I think the concept of metaphors is still super strong even for the phenomenal stuff because because you can't take any of it literally because you're not going to be the same later I mean you know as certainly for some animals it's more more than so than others but you you have to understand the Deep metaphor of what you learn not the not the specific details of your experience yes yes that's really good that's really nice there's a lot there um yeah it again it's converging with some of the stuff that I've been thinking about and writing a little bit in draft which is I'm kind of I'm coming to this idea that the self in some ways is a is a relevance filter a salience filter such that CU I think at a very deep phenomenal level everything that Finds Its way into attention or awareness is relevant it wouldn't really make sense and so I I would say that the one non-metaphorical thing there although you have the higher order concept of food is relevance and relev and meaning making to me and maybe well I I it's interesting you say you know we're trying to go up in abstraction up the generative hierarchy the one thing that the butterfly and the Caterpillar share is the imperative for nutrition and and maybe that's maybe so when we can start linking things in some kind of map of similarity maybe the thing that unifies them is preferences um so the caterpillar and that

maybe links us with the caterpillar and the Butterfly right we have this unified um goal that said the thing you said about the fact that they return to the same kind of leaf is very interesting because there does seem to be a idiosyncratic maintenance of memory uh which I can't quite account for there it made me think of Andy Clark's paper um knitting your own Markov blanket where he talks about the um metamorphosis of Markov blankets and how we can kind of get around this problem of like when is one thing another thing by adopting a process ontology rather than a substance ontology but friend or Mike you know whoever's got an idea on this is it is it possible it's just a false move to say that the butterfly and the caterpillar are the same thing like that's the same organism we not just say that you know there actually does become this division or this separation at and we it's philosophically vague as to when it happens but there are you know there was one thing and then now is another so so so I think the way uh the way to avoid these kind of philosophical traps is by being very practical and asking why why do you need to know so so if the right so so if the reason you need to know is is because you want to predict how much crawling around there's going to be then it's absolutely two different things because there's going to be zero crawling around afterwards but if but if what you need but if what you're trying to predict is um so what color substrate will it like to feed on then then then then you need to understand that it's the same thing because you will have less predictive power if you don't understand that it keeps memories so it's just again I think we're back to again the the the um the perspective of of uh you know what what question are you asking as the Observer yeah okay cool let's let's park that there I think something that you've worked on Mike but also France you mentioned and I want to pick up on is the difference between neural and non-neural structures um there's a there's a kind of neural bias in Psychology Neuroscience a quco bias um France one why is that the case do you think and two why should it not be the case if it should be the case maybe it should be the case yeah I mean one uh human Centric because we are humans but more more fundamentally the reason that that that we we are so confined on the idea of of neuros circuits being the only one that doing any inferences because they're just very fast right Action potentials are fast it's a highly optimal optimized Network optimized from what right neurons didn't come from nothing Asos didn't come from nothing electric potentials were were being used by cells way before the U innovation of of neurons so that is I think the reason why now we're trying to look at what by a neural organisms and just cells of of having these same properties um and more and I think other than just that being kind of transition what's also really interesting is Mike and I both it's one of my favorite um past daytime activities is asking neuroscientist what

what about your theory is specific to neurons and it's a question that neither of us has have gotten a good answer to yet because a lot of it of course you know there are certain certain time periods it takes if you're looking at fMRI imaging looking at the the oxygenated blood flow in the brain so you have to know those those time constants how fast does it take if you're looking at if you're actually measuring an act neuron you need to know all these transition times to really make your model take but if you change those time constants and you compensated for that in your model would still work so as far as we know um so far we're happy to get proven wrong on here but so as far as we know there's nothing that specific about neurons to that you need to make any of those models including influence actually work they highly optimized and they have completely different behaviors of course um because those timings do matter but they're not necessary for it because they're not NE they're not influence framework right it's not necessary to have a brain to have a model blanket it's not necessary to have a brain to have a generative model so once all the actual assumptions that go to a model can be applied to any robotic systems then you're good to go and you should apply it to those systems as well yeah I guess another biased mapping here is between cognition and the Brain um and there's I feel like there's been a a you know a movement towards obviously in terms of embodied cognition and the kind of dynamic the agent Arena Dynamic that was happening but also more recently might your paper with Anna Chong uh the brain is not mental you sort of I think you look at the immune system and sort of say how the immune system contributes as well as the neural system in subserving um adaptive control self-organization perhaps you could just speak to that and and say you know well I mean it's a it's a guess it's a fault experiment what would it look like if we didn't have the immune system if we were just trying to self organize purely in terms of neurodynamics yeah um I I mean we certainly there are uh there are knockout mice where the immune system is taken out and all that kind of stuff so and and a lot of self-organization does happen before the immune system kicks in I think you know I I think the idea is simply that uh there are many many different kinds of systems that go into this Collective that that we see when we say it's it's it's a human or or whatever and and all of them are potentially important contributors to the to the process um as as France said you know uh it's it's it's a really interesting experience to uh to ask people what a neuron is and and and watch them list things that are basically every cell in the body does this and and then and then you know so so I have another thought experiment that I sometimes do is like like back in the um you know back in the day when uh when the idea of neural networks was first being worked out right um I wonder if if if we had come along to uh to some of the

some of the greats uh in that in that field um and we said uh oh by the way just FYI um the biologist got it wrong thinking is in the liver not the brain so so there's two there's two ways they could have gone right one way is they could have said I don't care what you say we've got model of cognition and they require it to be to be neural so you guys must be wrong or they could say fine what do we care are are this what you know the the the basic calculus of you know mullock pit all that stuff the basic calculus of it doesn't require doesn't care what it is so fine it'll be right so so I I tend to think it would be I tend to think it would be the latter you know I don't see really anything in the in the Deep In The Deep lessons of um behavioral Neuroscience that's really about neurons per se you know it's about multiscale control it's about active inference it's about all these kind of deep things that go far beyond brains and they're applicable to all sorts of stuff stuff well are there certain processes which are reserved for the brain I mean I think someone listening to this probably a psychologist or a neuroscientist will say well take the brain out good luck take the liver out you know I mean still good luck but you might have a better chance thinking right uh for for a short amount of time I mean take the take the mitochondria out and uh and and you're done for pretty much at the same rate as as the brain now look the bottom line like I nobody's arguing that the brain isn't unique and interesting in certain ways right like we F you know as much as we find uh protocognitive capacities elsewhere I've never made the claim that language or longform planning is like we've not found it could be you know I wouldn't be terribly shocked if it were found but but we haven't found it so we don't make the claim so clearly brains are interesting and they do they do interesting things but um I think it's important to uh make make a distinction between between having found unique features and told a very specific predictive story that that why of why that is versus just this generic gut feeling that everybody has that that I have real you know real feelings and plans and this thing is a slime mold that I mean I don't know how many times people have said to me well that's a slim mold that's not a real decision it's just physics and and it just it drives me up a wall because what do you I mean you don't think we could tell a physics story about you like of course we could it wouldn't be a very good story but but neither is it about the slime mold and the fact that you can tell that story doesn't mean anything there's always a physics story to be told about anything and so you right I think I think it's it's fine to have specific theories about what it is the brain that's doing differently than other structures that's great um but a lot of the resistance to the kind of Continuum views that that France and I are talking about comes from a kind of closet dualism it's people really you know they just really feel they're different and and and they have to accept the fact that the

physics are not different so well now what is it then right it's like this then then there's something else and I think not not even that I'm particularly against that I just think it should be like come you know dragged out into the light and let's just say like say what you think the difference is you know I think I think that's really critical see I'm not sure the difference I'm not sure the jism really fundamentally is a uh brain body jism I think it's an awareness body dualism or Consciousness I mean like a traditional cartesian dualism I think is really at play here it's just we're we've been persuaded that Consciousness is rooted in the brain and so once you make that Association then you go well obviously my brain is special because it generates Consciousness Consciousness is special I can in principle imagine myself being conscious without my liver my leg my arm and so on and so that's the special part yeah is am I what am I missing there no I I think you're right but but I think I think I think what people are missing there is when they say conscious of my liver that isn't that isn't the question here the question so so right now your left hemisphere and my left hemisphere are having a delightful discussion about why Lang you know why Consciousness is in the brain and uh and and how important the brain is for for Consciousness that's fine then I will say to and I don't have a new Theory Of Consciousness by the way so I'm not like I usually don't talk about Consciousness at all but but since we're here I'll I'll just say that um I I have said that for the same reasons we think that the brain can support Consciousness we ought to take seriously that other parts of bodies also support Consciousness now at that point people say exactly what what you just said which is well I I don't feel like my liver is conscious but but that's a huge mistake of course you don't but you also don't feel that I'm conscious right that that that's that's that's exactly right so so so it's nice that the left hemisphere is eloquent and it has language and it can tell stories to to other left hemispheres about how how you know how conscious it is but that doesn't mean that what you don't have spread throughout different regions of your body is a non-verbal non-linguistic uh Consciousness that is not for us to access anym than we can access each others and that's that's the you know the fact that you don't feel it is no surprise there's no surprise there um you have to take its perspective and ask what do you think that's going on in your brain and so so literally um a student and I have have been uh we have this um we have this table that that has rows for the main theories of Consciousness and for each Theory there is something specific you know so so so Stu Hammer off will say it's microtubules somebody else will say it's you know electromagnetic field somebody else will say it's you know integrated information like what whatever it is and then and then we just look and say okay given that where does that occur in the body and the answer for almost all

of them is everywhere and so none of these theories as far as I can tell can distinguish you know uh and then and and some of them try to rescue it so right so so IAT has this postulate that says okay but but but there's there's we're just going to say there's only one right that that's that's that's a postulate you sort of add on to things right that's not so so that's that's the issue is that it's not about you feeling your your liver being conscious it's the first person perspective which may or may not be there yeah there's plenty to that I actually yeah I'm reticent to dive into Consciousness because all my podcasts end up being about Consciousness um and I have biologists on I don't want you know but yeah it's it's it's interesting I mean obviously a lot of this is being we have to take as a kind of um Axiom that at least for this conversation that Consciousness is uh well generated in the brain whether that's in a property je list or whether it's in a productive way and I also do just want to give a shout out to slightly more esoteric forms of idealism conscious realism and so on but it also makes me think about let's if we just get away from Consciousness in terms of first person perspective and just talk about basil intelligence in in in terms of morphogenesis it also makes me think well it's this is a term for Consciousness but people will talk about being whether conscious can be is multiply realizable has multiple realizability I.E if I put you know you know CH you know if I had a Chinese Nation their blocks idea with exactly the same functions as my brain exactly the same synapses because there's the same you know those connections would be the same between the members of the Chinese populace and my brain would I have Consciousness okay if we get away from consciousness I guess my question is is morphogenesis itself can it be is it multiply realizable in the sense that we're now talking less about the human body is there something sort of or and more towards like cyborgs or robots what you know people have this intuition I think as well that there is a life force and a l vital that leads maybe to this self-organization right there like there's got to be something there like or else why why are these things resisting entropy why are they self-organizing and I think the idea is like well if I just stuck some St of Prior in a robot why on Earth would that end up being self like why would that work to use very reductive language friend I'm just curious is there like an added ingredient at least it might be a metaphor but an added ingredient that we have to put into a non-carbon based thing in this case to to give it the chance to self organize yeah probably I would if I had to name be Evol evolvability um that would be one one part so if if the and it doesn't have to be evolution of of its actual physical components that can be evolution of its of its programming as well but that would be an important important point to it um the one thing I like to point out in for that question is right I I talked a lot about in this podcast about one my one of my biggest interest

is really understanding of where that drive for for an agency comes from but that's not just a kind of a philosophical um interest kind of it's actually really once once you find sufficient evidence for that being a fundamental Drive in evolution then you have to accept the the to combine that fact with the idea that well if any part of your body or any any robot um that you're putting into this is going to strive to maximize its agency with respect to its environment that is the real reason why if it turns out the liver is not conscious um right now then the answer would be if you accept the fact that well the drive is fundamentally in evolution any any life system is going to try to maximize its predictability and its agency over its environment then the answer would be well its environment isn't very interesting so the liver probably didn't have any any reason to maximize that drive if you were to put the liver in or you were to put a robot into an environment that's very active very much changing very volatile and you give it enough either enough time to evolve if you Comm to vulnerability or you just put in um a high degree autonomy over its own beliefs then over time with that driver it should be able to to to maximize that the problem right now is that the way we program robots is uh is very much right it's it's an very clear algorithm we don't allow any adaptiveness to it um and so that's that's how you would have to go about it I think and you may not want to do that just FYI but because the problem right is um you know we talk about one of my favorite um uh examples of why what difference between machines and and a brain is that a calculator is infinitely better than you at calculating you know adding numbers together multiplying it I mean you know and it's the dumbest thing ever like it's so simple and we've had these things for for decades now but so if you want to have a predictable reliable answer it's always correct you know the the human is not the best best way to do this that is the reason why we're not even if we had the theoretical framework to make a really agentic um intelligent system um for what we want them to do which just really make our lives Easier by making the same thing every day reliably you wouldn't want to do that but if your goal was to create a new uh intelligent um um cognitive agent then you would have to go about it differently yes that's an interesting little Paradox isn't it I mean like the most robust thing in the world might be a calculator but it's s quite useless it's like the more useful you become the more uh precarious your existences um Mike I was wondering if you had any thoughts on that kind of multiple realizability question and I guess what I'm going at with that is it's a very it's a it is a philosophical question but I think it's interesting for biologists which is why would Evol like why would Evolution prioritize self-organization I mean there just a fundamental question there in terms of entropy I mean why do we have these pockets of neg entropy entities um is

there a biological answer to that Beyond just well to be a biological entity you are neg entropy is is there anything can we get anywhere beyond that yeah um a couple couple of things I think are interesting and I want to be careful here because I I did I did a little while ago I did um I started writing a paper on what exactly is missing from our technology to provide true agency because I think we can actually say what some of those things are and it it quickly sort of dawned on me that to whatever extent I was on to something that would actually lead to a massive creation of of new beings with that that matter in the moral sense and not I'm not too interested in in you know being responsible for that so some other people will do it I'm sure but but but but I don't I don't want to do it so so so um but but I can but I can say a few things um I think I think uh when we I I I when when we make the distinction between life and Robotics um even aside from all the hyds and cyborgs that that show us that it's like a Continuum but even aside from all that when we make that distinction here here's what I hear I think life is what we call things that are good at scaling up uh goals so so you know when you have very when you have a bunch of subunits that have very basil competencies you know may maybe all they can do is follow least action laws or like like really simple things when when when you've got a pile of them and the pile has basically exactly the same cognitive lyone as the comp the components we say well that's not living that's a rock that's just you know but but but life is what we what we call things where there are um special I I I've also called it a cognitive glue there are like these special policies by which the parts relate to each other such that the collective has a bigger cognitive lone in new spaces that the parts couldn't have and that's and that's what we call life now now under that under that way of thinking about it could you make that out of other things I'm sure you could and I and you know I tend to think that in the wide universe I tend to think that there's probably lots of extremely diverse examples some most of which we wouldn't even recognize of of of that happening and I think um specifically I think I think what happens is that uh the The evolutionary process really I I I don't really think it makes specific solutions to specific problems the way that we do when we do genetic algorithms and things like this I think what what biological evolution does is it makes problemsolving machines and that's because if you overtrain on your evolutionary priors and you really take seriously uh what expectations that you might have about what genes you have what the environment is like if you take that too seriously you're done for because because guaranteed and this is why I also think that regeneration is not really about repair of um of physical damage like like external damage I think fundamentally regeneration is about knowing for a fact that your own parts are going to change right EV Evolution just

means the fact that you have a so some kind of a lineage that's subject to mutation whatever you you know your stuff is going to change if you don't accept that you're you're never going to survive long term so so what ends up Surviving are architectures that in particular are these multiscale architectures where the individual components have agendas and then this you know they they make up systems that have other agendas and and and on and on our our current robotics by and large is very flat right so you hope your robot is intelligent but the but the parts it's made of are not they don't really tend to do anything on they don't have their own agendas and I've given a talk called that why robots don't get cancer and this is why right it's because you they're made of parts that are not that that don't have their own little little goals in other spaces and there's no danger of them defecting unlike biology where it happens all the time so um I think what's I think what's happening here is that we end up with these architectures that uh are by virtue of cooperation and competition of parts that that end up being uh being having a CO a larger cognitive lonee you end up with this kind of this kind of intelligence ratchet and we could you know I could I could tell you stories about how that that type of thinking explains you know planarian the the amazing facts of planarian regeneration and so on this this this idea that what evolution I think really selects for is the plasticity to and and and it works out in different types of animals to different degrees but but overall the plasticity to deal with novelty you know this idea that when you when you emerge into the uh into the world you don't know for a fact what you are what you have is your genome the same as it was are your cells the rights shape or size do you have are you in the same environment you don't know any of these things you you have to um do your best in putting together a model of what that that that's why life is so interoperable that's why we can make anthr robots and xenobots and interface living tissue with crazy nanomaterials and make these Hots that are you just sort of totally cobbled together it's because by and large it's already bought into you know the the life we see today has already bought into the fact that um like like you know everything changes you know and you and you can't you can't depend on these things you have to figure them out in real time yeah yeah great it you know when you mention the word cognitive glue it made me realize that there's one thing that we haven't even touched upon at all I haven't even mentioned which is electricity and bioelectricity um so I think actually you had the paper mik didn't you last year two years ago um about bioelectricity being the glue of the cognitive mind soort of from physiology all the way up to the mind and obviously this is very much you know a seminal piece of your work is the processes of bi bioelectrical processes in cells and outside of cells in terms of self organization and I think at least to my

very Layman biological is electricity almost has the the kind of center of aam v a kind of thing that you know animates things and maybe you know I'm thinking about Frankenstein um either one of you I don't mind to everyone start what you know actually friend it was something that you mentioned which is maybe one of these ways that we can conflate the neural nonneural divid is the fact that both of them have both of them fundamentally electrical what what is it about that electrical ontology that animates or my animate intelligence or morphogenesis so when I was writing my my PhD thesis I was trying to answer the introduction the question of where does where does neuron even come from and the the best paper I found that was answering that question was looking into this very small organism that had just bunch of vdia around its surface it was a marine organism that was living on the bottom of the ocean and it was trying to Center environments and it by small changes in electrical fields that that were happing around it and that was causing um like many I won't say action potential but it was causing basically many vibrations and because us that to change change his movement so the reason I think why that would you know why electricity is that so important here is just very it's very simple mechanism to have that is physically a lot more accessible to all parts of the body that's actually the primary reason why I joined Mike laab and why I'm I've become a biologist I was a physicist before so I was looking for something in biology that would be all these would not be subject to completely different mechanisms if you switch organisms so like the moleular pathways we are typically investigating as biologists they are of as as they were before they are now still very important in all these different functions of the cell and organisms there kind of somewh unsatisfactory for the physicist because it's just so messy right you can't really make heads or tails of it and there is no clear field of potential that is is just one number essentially right you want to dce it to something very simple the beauty of using bity is that's you can you can have one number one potential you can like a me potential you describe to cell it's still made up of all the individual Parts there's many iron channels involved in this all the molecular pathways are still making up they all feeding into that one um one number that one B potential but what's more interesting is the red part right you saw if you have to constantly compare a millions of different molecules in a Cell that's not going a good good formulation for for achieving Consciousness for even doing any smart Behavior having something that you con constantly having a surface almost like a space that can read they can write onto and read out to I think that was makes electric potential so appealing that the only question then is like why wasn't it other physical Fields like mechanics they are of course very involved as well in developmental

biology and regeneration as well but mechanics don't have this neat feature where it's not as fast and it's not as localized so basically the way I see it the potentials have a beautiful tradeoff between being physical having a broad readability on on a cell or even a tissue level but are confined enough that they can be modulated enough to actually form active Behavior which you can't say for most for the other physical fields that we would um think about in this context great Mike you got anything you want to add yeah yeah a couple couple of things one one thing to add is that if if you think about um one of the really powerful things about um bioelectricity is that imagine an ion channel is specifically a voltage sensitive ION channel so as soon as Evolution discovers a voltage sensitive ION channel what you have there is an element with historicity it means that uh the current state is a function of what was going on before and it means that what you really have is a voltage gated current conductance AKA a transistor and so so so you know we know once you have that then then you can do anything and so uh electricity is is really good for these kinds of feedback loops right both positive and negative it's good for um representing information across time and space it's good for functionally integrating a system uh across across distance some some of the um really cool early uh evolutionary data on this come from bacterial biofilms and this is um G so's amazing work uh what basically he's he's got a paper called them you know brain-like signaling in bacterial biofilms and the idea is that already in a in a mat of bacteria so like a really long time ago um Evolution already uh HD onto this uh to this idea that um by using Electrical uh electrical signaling to integrate activity among competent subunits you can get gains at the collective level so they take turns it's used for for nutrient sharing and and things like this but but you get these Collective Dynamics and so I think it's just it's it's a very convenient modality for it that was that was found a long time long before neurons you know picked it up um it was used for that kind of thing but but no doubt there are other there are other ways to do it somewhere out in the universe there there there's other I'm sure I'm sure there are other kinds of cognitive glues out there but but here electricity works really well cool and so I know we need to wrap up relatively soon um and I guess it's it's a it's a kind of dark pun but us wrapping up kind of soon makes me think about aging and death and Decay and and whatnot and there's a there's a massive fuss about aging and Longevity at the moment um and David Sinclair and this notion of epigenetic scratches and it kind of makes me think of the inverse of that which might be regenerative biology and the capacity to reestablish self-organization reestablish uh the kind of priors that we've been discussing in terms of the functional priors the the preference priors the ones that you know are seeking a certain sort of uh external state or

seeking a certain probability distribution so as to be able to fulfill its function so to speak um what Mike what is the kind of lay of the land in your world as to aging um how seriously should we take this kind of epigenetic approach I won't ask about the moral aspect of it because it's a completely different question about whether we should live longer or not but sure you know having localized uh morphogenesis and self-organization is really the currency of biological systems how viable is it that we can reest that in something which is you know where self organization is ailing and and rather feeble yeah um I think well the the first thing is I I do just want to say something about the moral question because it also gets to the issue of regenerative medicine more broadly uh and uh and and Beyond regenerate medicine human augmentation and all of these kinds of things so so the one the one the one thing I want to uh point out is uh if it were the case that we were in some way optimized by some intelligence that shared our values then one could make the argument hey let's not push too far off of that right you're you're screwing up the planet and I get a lot of these emails from people saying uh well well things are great you bet you know you scientists better not mess it up and I just want to be clear that there there is no basis for for this for this view we our our life our health span our aging our susceptibility to to all kinds of dumb diseases and our lower back pain and our stigmatism all this all this stuff is is not in any way uh o optimized for any of the things we care about that's just where Evolution happens to have dropped us off at this point that's that's it and so there's nothing magical about this lifespan if if if you do you know people who are worried about um augmenting this lifespan well what if it had been 20 or 30 you know then what and you say well you know that's too short why is that too short it could have been anything it's but my point is it's completely arbitrary and so and so let's not um base our decisions about what um what ought to be on where the process of random mutation and selection has has dumped us so that's you know I kind of I think that's important to say um be beyond that I mean we're certainly involved in uh some some uh some research on aging and the role of bi electricity in aging and kind of the uh uh the degradation of bioelectric patterning information with age and all that but I think more importantly is the my my Weather Vein for all of this stuff is plaria so so I think you know plaria basically I think hold hold the answer to all of life's big questions and so uh I mean the the asexual form of um plenaria that we have they do not age they they they there is no there's no evidence of an old planarian they simply they they can go forever so these ideas that it's some sort of inevitable accumulation of errors or it's tiir or it's you know uh clearly there are ways around it that's the one thing that's we we have an existence proof uh that uh and then there's a few other animals

that do but planaria I think are the most interesting uh there there is obviously a strategy that gets around it and I think it's on us to figure out why that is and I think it's absolutely not a coincidence that planaria are also extremely regenerative in addition to being Immortal they're extremely regenerative they're cancer resistant and they have an incredibly noisy genome because they don't clean their genome in every generation like we do they right somatic mutations just if you if it doesn't kill the cell it just basically expands into the Next Generation BL can be mixed supposed every celllet have a different number of chromosomes they're just a complete mess and and those are the animals that are that are highly regenerative cancer resistant and Immortal isn't that amazing isn't that right yeah and and so there so so finally after after years of being like feeling scandalized about this you know the importance of the genome and all that and and why is the animal with the with the craziest genome also have all these amazing capacities um I think we actually finally have have a lead on on what's going on there but um but I think think I think that's that's what we need to look at in terms of Aging is is I I think if regenerative if the if the question of regenerative medicine were solved aging would be solved as a byproduct excellent yeah I mean it makes me think um in some ways I I don't know about the the function of planaria um you know that sort of the worms people can search up pictures of planaria um but it makes me think that they like the in many ways the there just take ourselves for example what makes a human a human is in some ways the fact that the the preference prior that we have or our limited functional our functionality is limited right like I am not going to be at 39 38 37 and 36 degrees Centigrade with equal probability but it sounds to me like planaria able to sort of metamorphosize in some way some of their preferences because as you say their genome is shaping is that is that kind of a misconstrue of what's going on is it that with or that with the sort of multi the the multiplicity of the genome the messy genome comes a multitude of function or is it that the function is still constrained yeah I'm I'm going to I'm going to give you uh and and and unfortunately have to we have to run in a couple minutes but I'll just give you a very quick uh story of what I think is going on in planaria uh I think uh and and this is um Steve Steve Frank told me an interesting analogy to this apparently uh you know when when raid arrays became popular right so computer um computer discs that are that makes it make copies of themselves so that you know if something goes wrong he basically said that when raid erase became popular the quality of disk media went down because it wasn't as important anymore to have to have your Hardware be super reliable because you've got this this the system on top of it I think what what's happened in planaria in that lineage is that it has uh really uh come to grips with the fact

that the hardware might be junk and that all the evolutionary effort went into cranking the algorithm basically the idea that we we're going to assume from the beginning that you're full of mutations you're unre it's it's unreliable hardware and we're going to assume it's unreliable and so what we're going to do is do our best to have a system that is so highly regenerative that no matter what's going on we can we have St a pretty good chance of making a good planarian the the the development of planaria is called a chaotic mode where it's a total mess and then it sorts itself out into a proper worm um I think uh and and here's why I think it happened and and we have computational models that show that show how this works imagine that you you have a uh certain competencies to fix defects for example like in the Frog we know that if if the mouth is moved off a little bit in development it'll soon make its way back to where it needs to be it's sort of fixed fixes itself so so in evolution if you have a system that has a little bit of Competency to fix itself when that individual comes up for selection and selection sees that okay this is a great individual what selection doesn't know is is it great because the genome was amazing or is it great because the genome was actually so so but it fixed itself you don't know once you don't know you're it becomes really difficult to reward for the best genomes what you can reward is for the best phenotypic fitness for the best outcomes and as soon as you do that that makes the problem worse because then the competency goes up then then it becomes even harder to know what your genome was so so you so you start off with this ratchet with this positive feedback loop and I think in planaria it went all the way and in another species not so far and it's some like *celegans* maybe not at all but but but that that idea that that the the more competency of of um uh kind of uh uh reparative regenerative developmental process it it it you can you can eventually if you take it to its logical conclusion you can say fine we're going to assume that the that the structural genome is going to be all kinds of noise and we're going to come up with an algorithm that can keep it clean keep the the morphology clean assuming that that's assuming that that's the case and I think planaria just like cranked that knob all the way and salamander sort of do it and and you know and maybe neems don't do it much that is very very cool it makes me want to be a biologist because it the question if we had more time and I know we don't is well where on Earth is that regenerative mechanism coming from if it's not encoded in the genome somewhere but we won't go into it it just sounds to me like there are other ways to maximize model evidence to use these active influence words to maximize model evidence or to return to an attractor State Beyond genomic stability and that's really fascinating because I think people think well the the priors the So-Cal phenotypic priors are just what evolution has

endowed Us in terms of our DNA yeah and and and so if we wanted to have another discussion this is where we would talk about xenobots and anthr robots and these things that have never been under selection at the organism level here you know in the history yeah I mean I've also been I'll finish up here I've been reading a little bit of Evan Thompson in mind in life and he talks about how well the the autopoietic coupled system is not going you can't discover that dynamic in the genome because there are things that are also selected over time which which refer to the external Dynamic and which refer to the world in which we are born and thrown into and that's really important as well and I wonder whether what that interplay means in terms of regeneration and and morphogenesis but as as you're all aware and I'm aware we're Limited in time so we should I'd absolutely love to do this again I mean it was absolutely fascinating um guys thank you uh just if you have a minute it' be great to know where people can find your details and also like what you guys have got coming up in the pipeline that people can be aware of France you want to go first sure I mean you can basically look at our names and and you know SC and get some of our papers um we have a great website the for 11 La there has a bunch bunch of our research come eliminated um my uh my papers coming up we actually look in some of these questions in an anual organism in volvox that's going to be happening soon I'm very excited about that uh because all all the works published for me right now is a lot of the the theoretical simulations and talking about the concepts but I'm excited about most of my PhD work was trying to apply this to an organism and see whether or not there's anything there um so that be fantastic and Mike yeah so so the lab website is at uh dr mike 11.org one word dr mike 11.org and so that's where all the peer reviewed stuff the the papers the protocols um data sets software everything is is there and then kind of a little bit more out there some thoughts are at a Blog called thought forms. lifee and that's just my that's just my personal one I'll make sure everything goes in the uh video description so people can check it out there but guys thank you again thank you so much yeah yeah great conversation thank you so much thank you it's pleasure